Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/01/2006
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Investigation of Future Microgravity Atomic Clocks		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	04/01/1998	**End Date:**	03/31/2006
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	5	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2905		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity.		
Task Description:	We are developing new techniques to significantly advance atomic clocks. The microgravity environment of the ISS can lead to significantly better atomic clock due to the potentially long interrogation times. RACE (Rubidium Atomic Clock Experiment) is a flight project that will use Rb atoms to avoid the large cold collision frequency shift. It will advance atomic clock science, significantly improve the classic clock tests of general relativity, and globally disseminate time and frequency. The objective of RACE is to build a microgravity-based laser-cooled atomic clock capable of better than 10-16 accuracy and a stability approaching 10-15 for an averaging time of 1 second. These will be compared against an earth-based clock to establish the clock's accuracy and also to perform tests of general relativity. Atomic clocks have many applications ranging from navigation to the synchronization of high-speed communication networks. Developing more accurate atomic clocks is important because they may enable dramatic improvements in interplanetary navigation and potential improvements to the widely used Global Positioning System GPS. The microgravity environment of the		

	ISS can allow dramatic improvements in the accuracy of atomic clocks and also provides a natural international time and frequency dissemination capability. A comparison of clocks on the ISS and clocks on the ground will lead to improvements as large as one million on precise tests of general relativity. In addition, our measurements of cold-collision cross sections show that Rb clocks can be much better than the accepted Cs standard that defines the SI second. The new techniques we have developed have advanced the capabilities of earth-based atomic clocks, future microgravity clocks, and other precision measurements in atomic physics. As a prominent example of the impact of our measurements, laser-cooled Rb clocks are currently under construction and will be the ground reference clock for GPS beginning in 2005.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on Rubidium clocks has lead to more accurate atomic clocks. As a result, a number of Rb clocks are currently under construction and these are expected to be the ground reference clock for GPS beginning in late 2005. Our work on juggling and microwave cavity designs has also changed the designs of clocks around the world.
Task Progress:	We have continued our work on future clocks based on optical transitions. We have locked two lasers to two independent high finesse cavities, achieving a beatnote linewidth of 23 Hz with a stability of 4 Hz for 0.1 seconds of averaging. We have demonstrated direct locking of a noisy laser to a high finesse cavity. Previously, it was believed that two successive cavities were required. We theoretically showed and experimentally demonstrated that the laser can be directly locked to a very narrow cavity when the laser sweeps through the cavity resonance in less time than it takes for light to build up in the cavity. This result is important for future optical frequency atomic clocks and also is relevant for experiments to detect gravity waves, such as LIGO and LISA where, during lock acquisition, the cavity will sweep very quickly through resonance. In our cesium atomic fountain experiment, this year we have seen the first signals for a new type of scattering experiment where we directly observe the difference of the s-wave scattering phase shifts. This may allow us to precisely probe the scattering length variation near a Feschbach resonance, sensitively probe two-body forces at short range, and provide vastly improved information about atom-atom interactions. Our work on modeling microwave cavities was presented in two conference talks. We have shown interesting power-dependences. Our analysis demonstrates that some current techniques used to evaluate the accuracy of atomic clocks need to be altered. The work will show the limits for distributed cavity phase shifts and has lead to improved microwave cavities for atomic clocks.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
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Papers from Meeting Proceedings	Ruoxin L, Gibble K. "Distributed cavity phase and the associated power dependence." 2005 IEEE International Frequency Control Symposium and Exposition, August 2005. In: Proceedings of the 2005 IEEE International Frequency Control Symposium and Exposition, p. 99-104, 2005. , Aug-2005
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